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Review Article

The analysis of ancestry with small-scale forensic panels of genetic markers

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In the last 10 years, forensic genetic analysis has been extended beyond identification tests that link a suspect to crime scene evidence using standard DNA profiling, to new supplementary tests that can provide information to investigators about a suspect in the absence of a database hit or eyewitness testimony. These tests now encompass the prediction of physical appearance, ancestry and age. In this review, we give a comprehensive overview of the full range of DNA-based ancestry inference tests designed to work with forensic contact traces, when the level of DNA is often very low or highly degraded. We outline recent developments in the design of ancestry-informative marker sets, forensic assays that use capillary electrophoresis or massively parallel sequencing, and the statistical analysis frameworks that examine the test profile and compares it to reference population variation. Three casework ancestry analysis examples are described which were successfully accomplished in the authors' laboratory, where the ancestry information obtained was critical to the outcome of the DNA analyses made.

Introduction: the context of ancestry analysis in forensic identification

Forensic ancestry analysis makes use of the small proportion of human diversity found between different globally distributed population groups — broadly defined by the world's continents (but also including South Asia, or the Indian sub-continent). Such emphasis on continental population group differences tends to underplay the much larger proportion of diversity between individuals within the same population group — amounting to some 85%–95% of total diversity [1,2]. Emphasis on human population differences, no matter how small a proportion of total human variability these represent, inevitably brings into play a wide range of ethical considerations. Notably, careful regard for the risk of targeting and stigmatising particular communities, with the burden of proof passed to a minority community's members by a police investigation obtaining ancestry data from crime-scene DNA samples [3,4]. At the time-of-writing, several European countries have opted to allow ancestry analysis in a forensic context, but others have decided ethical considerations argue too strongly against its use. Beyond investigative applications, knowledge of the likely ancestry of a DNA sample can provide useful information in other forensic fields, including: more detailed identification of missing persons; aiding triage of bodies in mass disaster victim identification; knowing the ancestry of samples held in short tandem repeat (STR), Y-chromosome or mitochondrial DNA databases, thereby ensuring their geographic accuracy; and, refining familial search systems that require accurate STR allele frequencies to correctly evaluate the estimated relationships [5]. Beyond forensics, DNA ancestry tests with the necessary sensitivity are more often successful in inferring the ancestry of medical archive material or archaeological samples [6]. Because forensic tests must be small-scale to maintain the required sensitivity to contact trace DNA, only genetic markers with the most highly contrasting variation between populations can be adopted. For this reason, the degree of geographic precision claimed by

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commercial suppliers such as direct-to-consumer (DTC) genetic test companies, is not a realistic outcome from forensic tests genotyping 3000-fold less markers — a maximum of 200 markers in one forensic multiplex vs. 700 000 loci in a typical DTC microarray test.

Some discussion about terminology is relevant, since forensic scientists prepare reports for the guidance of police investigators requiring complex DNA information to be summarised in meaningful, understandable terms. We strongly advocate the use of geographic descriptions of a person's ancestry, namely African, European, East Asian, etc. and avoidance of linguistic or cultural terms such as Hispanic, Latino. Both these labels are regularly used even in genetics studies, along with Caucasian to describe Europeans, although this is an imprecise historical term lacking validity. Likewise, the widely used term ethnicity has little relevance as a description of genetic ancestry (commonly termed biogeographical ancestry), which is what the forensic DNA tests we describe here measure. Biogeographical ancestry can be defined as variation found in an individual that can signal their origin from a particular geographic region.

Ancestry informative markers

Polymorphisms showing highly differentiated patterns of variation between populations are termed ancestry informative markers (AIMs) and are obvious candidate loci to combine into PCR-multiplexed forensic ancestry tests [7]. The vast bulk of AIMs adopted for forensic use have been autosomal single nucleotide polymorphisms (SNPs), supplemented more recently with insertion-deletion polymorphisms (Indels) [8]. Both marker types are predominantly binary polymorphisms, and when ancestry informative, show contrasting allele frequency distributions between populations. The statistical inference of ancestry makes a direct link between the observed allele frequency in a population and the likelihood an individual with the allele originates from that population [9], e.g. an allele at 0.6 frequency in one population, indicates a six-times higher likelihood for a person with the allele coming from that population, than from a population with a 0.1 allele frequency. SNPs and short Indels (with insertion-deletion alleles a few nucleotides in length) have the additional advantage for forensic tests, of being amplified in very short fragments compared with longer STR markers used in standard DNA profiling. Markers on short sequence fragments are more resistant to DNA degradation and therefore offer better success analysing DNA from challenging forensic samples [10].

Markers typed on longer amplified fragments, including routine or specialised STRs [11–13] and microhaplotypes [14] also offer ancestry informativeness in a small proportion of loci. STRs have the advantage of straightforward analysis with sensitive capillary electrophoresis (CE) detectors calibrated for routine profiling. Although specialised ancestry-informative STRs are less differentiating than SNPs [13] and have not been widely adopted, alternative use of routine STR data to analyse ancestry is sometimes the only way to infer ancestry when all DNA extract is exhausted — as is often the situation in cold case reviews. Luckily, routine STRs offer a practical level of differentiation between the five main continental population groups (Africa, Europe, East Asia, America and Oceania), with 32 autosomal STRs achieving <1% assignment error applying a probability threshold (i.e. only reporting probabilities of >10 times more likely a particular population) [11]. Microhaplotypes, comprising groups of SNPs with divergent allele frequencies closely aligned on short sequence tracts, are gaining interest since they can be readily typed using massively parallel sequencing (MPS). In a recent study of different kinds of forensic ancestry marker, carefully chosen microhaplotypes were shown to be as informative for ancestry as the best binary AIM SNPs [15]. As microhaplotypes are particularly effective at analysing mixed DNA, the possibility of inferring the ancestry of contributors in simple mixtures using microhaplotypes is now being explored [16].

Lastly, the widely used forensic markers of Y-STRs [17] and mitochondrial DNA (mtDNA, [18]) have long been used to infer ancestry from forensic tests where they are the appropriate DNA analysis choice: Y-STRs often used to identify male suspects in sexual assault cases; and mtDNA the system of choice for hair analysis (where nuclear DNA packaging in the shaft precludes conventional profiling). Both markers are strongly geographically stratified, and in the absence of recombination in each case, ancestry inference from their data relies on large-scale global variation databases, notably YHRD for Y-STRs [17], and EMPOP for mtDNA [19]. These non-recombing, uni-parental loci represent single markers, so an ancestry inference may be based on an atypical lineage which contrasts with the overall pattern of recent ancestry revealed by autosomal variation. The dangers of making a forensic ancestry inference from non-recombining markers were highlighted by the discovery of the African hgA1 Y haplotype amongst seven European males from Northern England with the same surname [20]. However, mtDNA and Y-STR or Y-SNP [21] data can help add extra detail to the analysis of

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DNA from individuals who have admixed backgrounds and show co-ancestry patterns in autosomal markers [22]. If the admixture is recent — e.g. parental or grandparental, the analysis of the paternal lineage with Y data and maternal lineage with mtDNA can more effectively elucidate the likely pattern of co-ancestry in the individual [22].

Ancestry assays applicable to forensic DNA analysis

The first forensic ancestry assays pre-date the DNA era, starting with serological tests for the Duffy blood group system used since the Sixties, progressing to include protein polymorphisms with African-specific variation in erythrocyte systems: Pep A; Hb S; CA II (now genomically described by SNPs: rs733062; rs334; rs2228063, respectively, with the Duffy SNP rs2814778 still the most powerful forensic ancestry marker - see Figure 1 in [7]). Once STRs became established, they routinely gave varied profile frequencies when applying different population allele frequency estimates (i.e. the likelihood of the detected profile in any one population

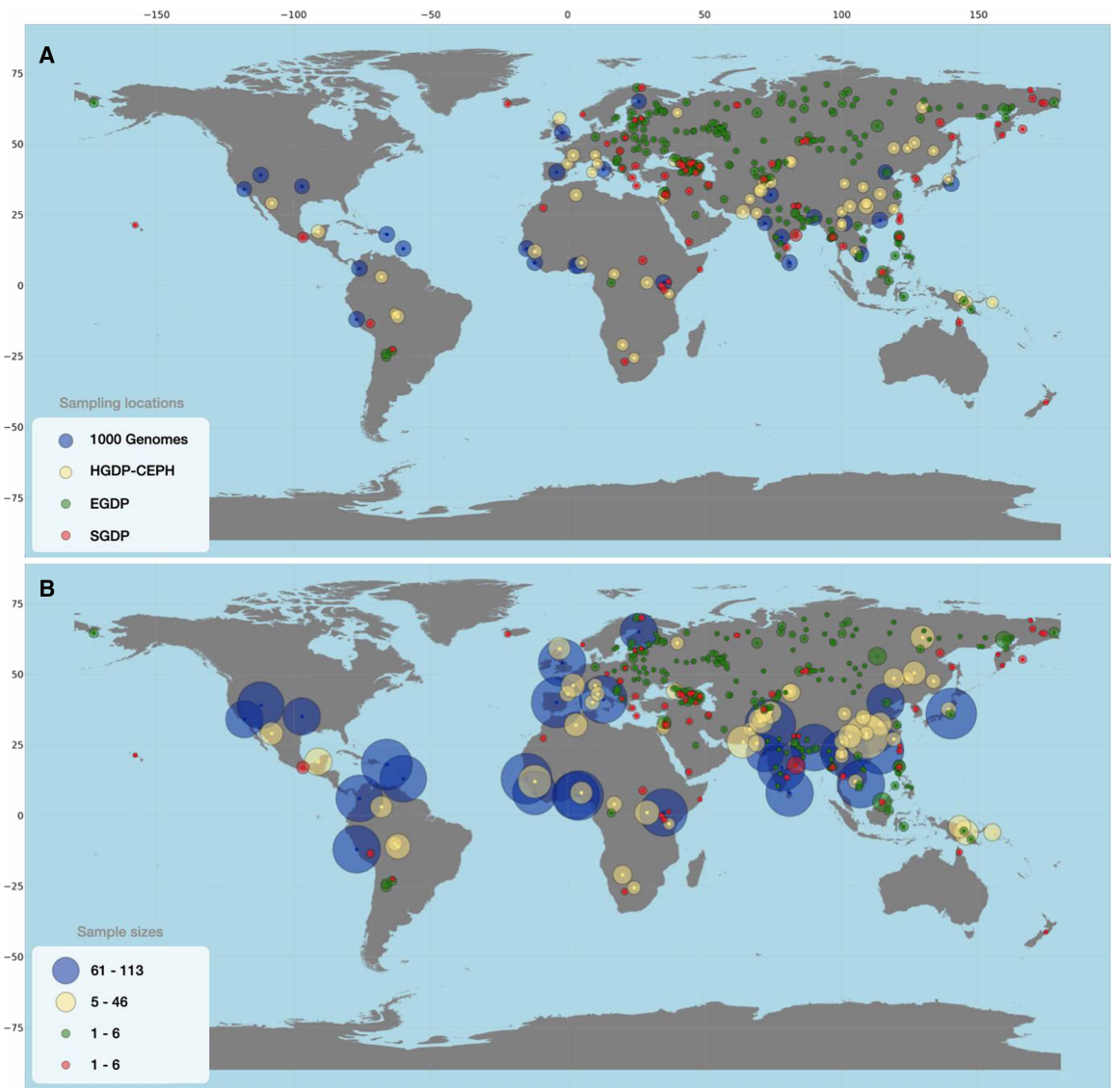


Figure 1. Sampling locations of population analyses performed by four major international whole-genome sequencing consortia.

(**A**) Circles show log sample size to minimise overlaps. EGDP: Estonian genome diversity project; SGDP: Simons Foundation genome diversity project. (**B**) The same sampling location data as (**A**), with circles proportional to sample sizes for each consortium.

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often differed by several orders of magnitude). UK forensic DNA tests routinely report profile frequencies for European, African-Caribbean and South Asian populations mirroring the current UK population profile. This practice led to the first systematic forensic ancestry analysis study of Lowe et al. in 2001 [23] looking at six-STR profile data. Lowe proposed a Bayesian approach: constructing likelihood ratios that compare cumulative DNA profile frequencies (whether a few STRs or several hundred SNPs), which now forms the core statistical technique for estimating probabilities of ancestral origin based on allele frequency reference data for a predetermined set of populations.

Since the emergence of SNPs as forensic AIMs of choice, two multiplexed PCR SNP genotyping techniques have been widely applied to forensic DNA: (i) SNaPshot primer extension tests [24] where assays of up to 34 multiplexed SNPs can be genotyped using established CE detectors, and (ii) MPS sequencing assays with much larger multiplexing capacity and enhanced sensitivity to low level DNA. Several state-of-the-art MPS assays dedicated to forensic ancestry analysis have been developed with comparable multiplexing scales: EUROFORGEN gAIMs (127 SNPs [25]); Thermo Fisher Scientific's Precision ID Ancestry Panel (165 SNPs [26]); MAPlex, combining 144 SNPs with 20 microhaplotypes [27]; and the VISAGE basic tool which combines 112 ancestry informative SNPs with 41 pigmentation predictive markers [28]. SNaPshot as a forensic technique is difficult to regularise and lacks consistency both between runs and between laboratories following standard genotyping procedures. The peak patterns generated also show considerable signal variability, making it impossible to efficiently detect mixed DNA. These factors prompted the development of a 46-plex ancestry-informative Indel test using dye-labelled PCR primers, which allows direct PCR-to-CE detection of amplified products in the same way as STRs. This innovation brought a more reproducible and regular ancestry analysis workflow compared with SNaPshot genotyping (see Figure 2), and the PCR-to-CE technique allowed 19 laboratories to successfully genotype these Indels and detect mixed DNA samples in a collaborative exercise [29]. For forensic laboratories without MPS technology in place, but having established CE pipelines, the 46-plex Indel assay offers the most straightforward way to instigate an ancestry analysis system [30].

Since 2015, the principal developments in forensic ancestry analysis have centred on MPS. To maximise the information per sequence obtained from MPS analysis, SNPs have been supplemented in two assays with ancestry-informative microhaplotypes. MPS multiplex sizes have also begun to be enlarged beyond combinations of 130–170 markers. Although ~80 markers are sufficient to differentiate the five continental population groups and South Asians, there is the need to distinguish Middle East populations from Europeans and South Asians (for the authors' laboratory, this distinction represents the most common operational request from investigators). It is already the case that differentiation of Middle East populations requires >100 markers, as individual AIMs for this analysis are less divergent than those used for continental population distinctions. The NAME panel [31] was developed to distinguish individuals originating from North African and Middle East regions and contains 111 SNPs, but only works efficiently when combined with the 126 gAIMs SNPs [25]. We outlined above that detailed analysis of co-ancestry patterns in admixed individuals benefits from the combination of data from Y markers and mtDNA with autosomal AIMs. The main goal of the VISAGE Consortium (http://www.visage-h2020.eu) has been to build single-assay tools for the joint inference of a crime-scene DNA donor's appearance, ancestry and age. Although age estimation using methylation analysis requires a separate assay due to the specialised preparation of the DNA, all appearance and ancestry markers are combined in multiplexes of 153 loci (41 to predict appearance) for the VISAGE basic tool [28]; and 226 AIMs (mainly SNPs) plus 194 appearance loci for the VISAGE enhanced tool (publications in preparation).

Reference population data

Accurate and representative reference population data is an important part of statistical assessments of the frequency of the AIM profile obtained from the forensic assay. Variant data from different geographic regions can be obtained through direct genotyping of DNA samples collected from the populations of interest using the specific ancestry assay. A more economic and productive approach is to compile data from publicly available whole-genome sequencing data published online by international genomic consortia. Such data is the most useful resource for both developing new ancestry markers and compiling reference data for statistical analysis.

The variation database ALFRED [32], contains allele frequency data from >700 populations and 600 000 polymorphisms. Similarly, allele frequencies can be compiled from the genome Aggregation Database (gnomAD) [33], which lists sequencing data from a range of large-scale genome projects. Note use of both databases is limited as most ancestry inference methodologies require individual genotypes as input.

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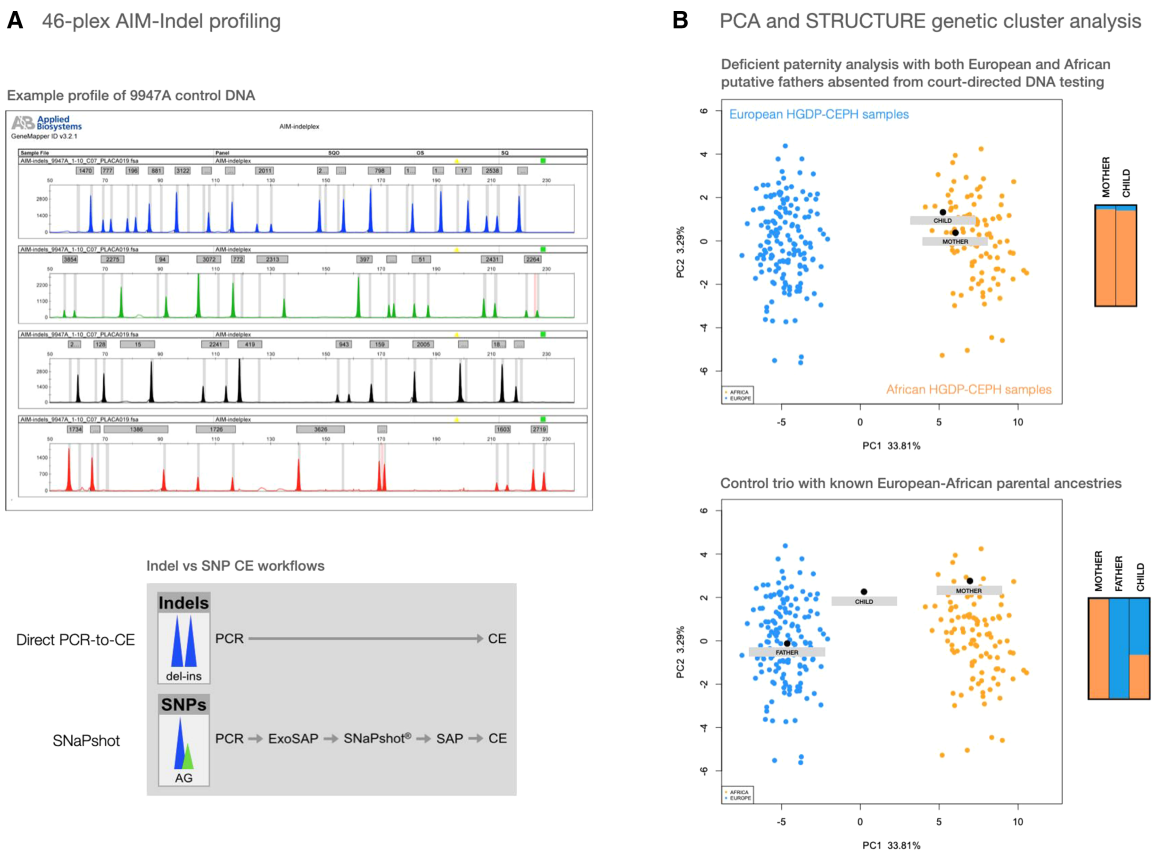


Figure 2. Using Indels as ancestry-informative markers in a simple kinship analysis.

(**A**) 46-plex AIM-Indel forensic ancestry test details showing a typical profile and the summarised PCR-to-CE workflow compared with CE-based SNP genotyping with SNaPshot. (**B**) PCA and STRUCTURE population data analysis performed to resolve a disputed paternity where both putative fathers had absented from DNA testing. Upper plots show the tested mother and child, lower plots show a control family trio, with a European father and African mother.

The largest source of individual genotypes is the Phase III data release from the 1000 Genomes Project, comprising 2504 whole-genome sequences of individuals from 26 populations, grouped into five 'superpopulations': African, European, East Asian, South Asian and Ad Mixed American [34]. Genotypes can be obtained with the Data Slicer tool from Ensembl Genome Browser [35] or as Variant Caller Format (VCF) files directly from 1000 Genomes (http://ftp.1000genomes.ebi.ac.uk/vol1/ftp/). The latest data release also contains high sequence coverage (30×) re-analysis by New York Genome Center. Alternatively, Phase I data [36] from 1092 individuals, is accessible through the SPSmart browser [37].

The Human Genome Diversity Project-Centre d'Etude du Polymorphisme Humain (HGDP-CEPH) panel [38] comprises 952 individuals from 52 worldwide populations [39]. Due to relatively small and variable sample sizes per population, individuals are often rearranged into population groups (equivalent to those of 1000 Genomes): Africa, North-Africa/Middle East, Europe, Central-South Asia, East Asia, Oceania and America. Data from 650 000 SNPs [40] can be accessed through SPSmart [37], while recent whole-genome sequencing data from 929 individuals [41] is available from the Wellcome Sanger Institute (ftp://ngs.sanger.ac.uk/production/hgdp).

Finally, two other whole-genome sequencing projects provide relevant population data for forensic analysis. However, they both have very limited numbers of samples per location (1–6 individuals). Therefore, these variant databases are more suitable as test data rather than reference data. The Simons Foundation Genome Diversity Project (SGDP) [42] has whole-genome sequenced 260 individuals from 143 demographically smaller populations, chosen to span as much human genetic variation as possible (39 Africans, 23 Native Americans, 27 Central Asians or Siberians, 49 East Asians, 27 Oceanians, 38 South Asians and 71 West Eurasians). Of these, 24 samples overlap with 1000 Genomes Phase III samples and 122 with the HGDP-CEPH panel, leaving

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114 SGDP-unique samples located specifically in regions not covered by 1000 Genomes: Central South Asia, Northeast Asia, Southeast Island Asia, Near or Far Oceania, and unadmixed Native Americans. The Estonian Biocentre Genome Diversity panel (EGDP) public data comprises 402 samples from 126 populations, focusing on the Eurasian continent and covering geographic areas in East Europe, Central South Asia, India, Mainland/Island Southeast Asia and, similarly to SGDP, including the Northeast Asian regions of Siberia [43].

The global distribution of 1000 Genomes, HGDP-CEPH, SGDP and EGDP population sample areas and shown in Figure 1, and more detailed geographic information has been compiled by Phillips et al. [44].

Statistical analysis to infer ancestry

Several statistical methodologies are applicable to ancestry inference and the most widely used of these can be grouped into three categories: methods based on dimensionality reduction, model-based clustering methods, and methods using Bayesian likelihood calculations. The first two classification methods are unsupervised, meaning the information linking the populations-of-origin of individuals in the reference dataset is ignored, while the third is supervised — where classification requires a reference dataset specifying population information.

First, the dimensionality reduction approach, including Principal Component Analysis (PCA), Multi-Dimensional Scaling (MDS) or Principal Coordinates Analysis (PCoA), reduces the complexity of data by condensing the largest possible portion of the total variability in geometric space. Analysis results are often represented graphically in two or three dimensions, with the axes representing the principal components (PCA) or eigenvectors (MDS/PCoA), and the individuals (usually coloured or shaped according to their population) clustered in Euclidean space. These analyses can be made with dedicated software, including R and MATLAB. Alternatively, the *Snipper* classification portal (http://mathgene.usc.es/snipper/) produces two-dimensional PCAs from binary SNPs or Indels. Assessment of ancestry from these methodologies relies on the visualisation and interpretation of the reference population clusters and the position of the unknown sample relative to the reference clusters. In this way, an unadmixed sample will cluster together with the reference individuals of the same origin, while an admixed sample will position between the reference population clusters of the co-ancestry components.

Second, model-based clustering methods, notably STRUCTURE [45] and ADMIXTURE [46], use iterative algorithms to distribute samples into a previously specified number (K) of groups or genetic clusters whose members share similar patterns of variation. Using these methodologies, a critical factor is the accurate estimation of the optimal K value without overinterpretation of the data. The resulting genetic cluster membership coefficients for the K clusters for each input sample are usually represented as bar plots in which each inferred K group (in different colours) signifies the range of ancestries in the data. This allows a straightforward interpretation of ancestry: unadmixed samples are mostly coloured accordingly to the corresponding reference populations and admixed samples show combinations of two or more colours at different proportions, corresponding to the co-ancestry components.

Third, methods based on Bayesian likelihoods calculate a probability of origin from each population of the reference dataset and infers ancestry from the likelihood ratio (LR) comparing the two highest probabilities. Both the FROG-kb portal (https://frog.med.yale.edu) and the stand-alone FrogAncestryCalc [47] calculate relative likelihoods of ancestry from different reference populations for several fixed forensic SNP ancestry panels and provide LRs for each population comparison. The *Snipper* portal allows use of custom population datasets and produces an LR between the two populations with highest probabilities. GenoGeographer (http://apps.math.aau.dk/aims/) performs a similar analysis, adding a likelihood ratio test [48] that provides guidance on whether the populations most relevant to the sample are included in the reference dataset, e.g. the test individual could originate from a population that is not included or is genetically distant from those in the reference set. For these approaches, high LRs can be interpreted to signify an unadmixed sample from the most likely population, while low LRs can be explained by admixture in the tested individual [49].

Finally, none of the above methodologies provide an absolute answer but requires interpretation from the forensic expert. Therefore, an approach combining several methods is essential to avoid possible misinterpretation of the results in those cases where a particular system has limitations (e.g. overlapping PCA clusters, admixed samples giving low or ambiguous LRs). At all times, the practitioner should be aware that the reference population data used, no matter how extensive, cannot be fully representative of many extensive geographic regions of the world. As an example of the interpretative risks that arise from a lack of sufficient genetic data, the analysis of single Polynesian SGDP Maori and SGDP Hawaiian donors with MAPlex, gave very different patterns of variation compared with the Melanesian reference populations used to make

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inferences across the whole of Oceania (see Figure 6 in [27]). GenoGeographer measures a *z*-score to attempt to gauge such disparities and alert the user when they occur [49].

Three cases exemplifying contemporary forensic ancestry analysis

A paternity test resolved without testing the two putative fathers

Application of ancestry analysis to paternity testing is unusual, but this case illustrates the simplicity and speed of Indels in accomplishing a straightforward continental population group differentiation, and detection of co-ancestry when critical to the interpretation of results. Figure 2 shows a typical 46-plex AIM-Indel profile achieved after ~3 h PCR and ~40 min for CE, with summarised single step vs multiple step workflows for PCR-to-CE and SNaPshot, respectively. The court examined a paternity claim by a South African woman on two putative fathers, one from South Africa, one from Europe, but both had absented themselves from testing. On the presumption that no other putative father was claimed, the court accepted that the ancestry analysis of the mother and child's DNA excluded the European man. The PCA and STRUCTURE patterns of Figure 2 indicate unequivocally that both mother and child have unadmixed African ancestry, in contrast with a co-run control trio with a European father and African mother. The child from this trio gives a PCA position midway between the parental cluster positions, and STRUCTURE gives near identical (50 : 50) coefficients of cluster

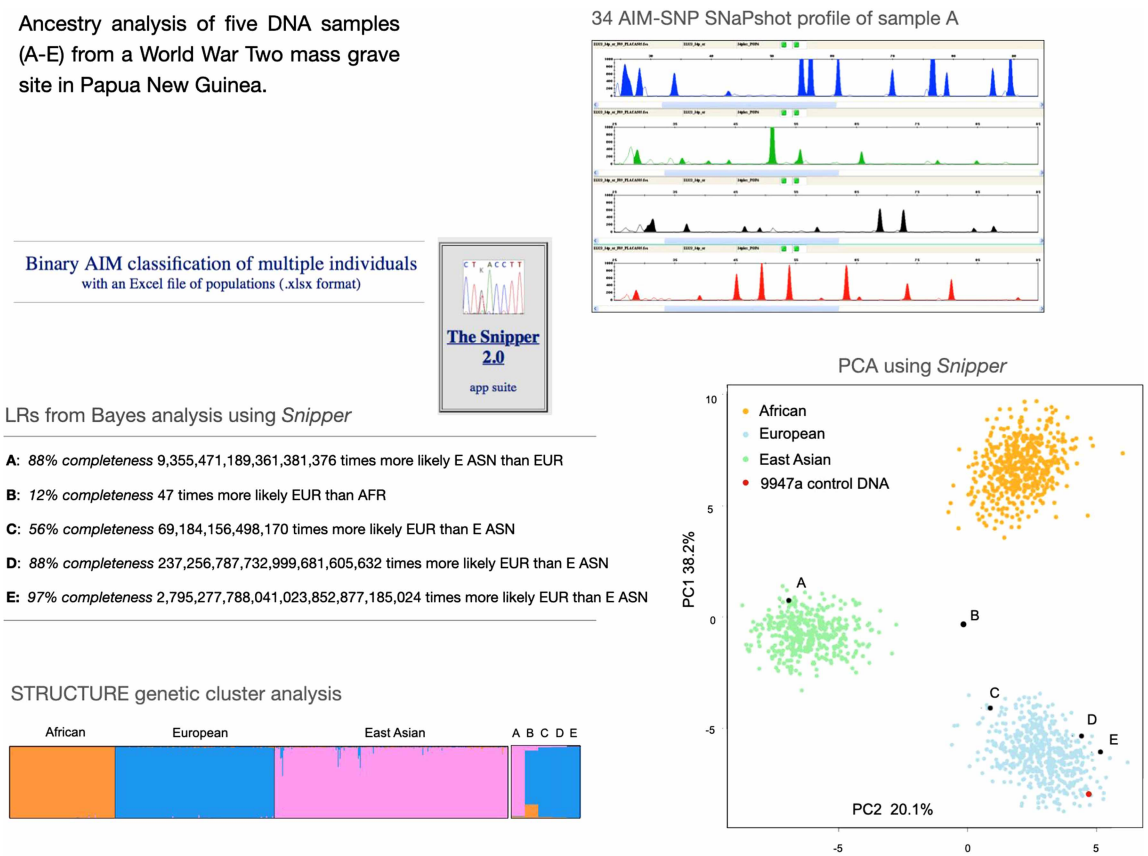


Figure 3. Bayes likelihood tests, PCA and STRUCTURE population data analysis using 34 AIM-SNPs in a CE assay, applied to a Second World War soldier identification program examining a mass grave site in Papua New Guinea.
Results indicate four of five skeletal DNA samples provided sufficient information to identify their ancestry and could be re-patriated for burial or further identification tests. Sample **A** was inferred to be East Asian and samples **B**, **D**, **E** were inferred to be European. Sample **B** had insufficient SNP data for a secure inference to be made (only 4 of 34 AIM-SNPs successfully genotyped), indicated by inconclusive PCA and STRUCTURE results, as well as a very low likelihood ratio from Bayes analysis.

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membership in a K : 2 analysis. As a rule of thumb, pairwise population comparisons usually produce much more clear-cut results than analyses of multiple possible ancestries [50].

Identification of Second World War soldiers at a mass grave site

In this example of ancestry testing applied to disaster victim identification, ancestry inference was used to help triage the remains of Second World War soldiers from Japanese and Australian armies, who had engaged in combat in Papua New Guinea and were consequently buried together at the battle location. On-site triaging was important as Japanese remains were re-patriated and cremated without further identification steps, whereas Australian remains underwent analyses to establish matches to surviving relatives. Figure 3 shows typical AIM SNP analyses made using a 34-plex SNaPshot assay with a proven track-record of success with highly degraded DNA [10] and highly efficient at differentiating African, European and East Asian population groups [51]. In the five analyses shown (A-E), there is a clear relationship between the LR values obtained from *Snipper* and profile completeness. When 19–30 of 34 SNPs are genotyped successfully, the LR values and PCA positions give reliable inferences of East Asian (A) and European (C–E) ancestry. Viewed in the absence of its genotyping success data, sample B could be inferred to indicate co-ancestry in the PCA and STRUCTURE patterns, but

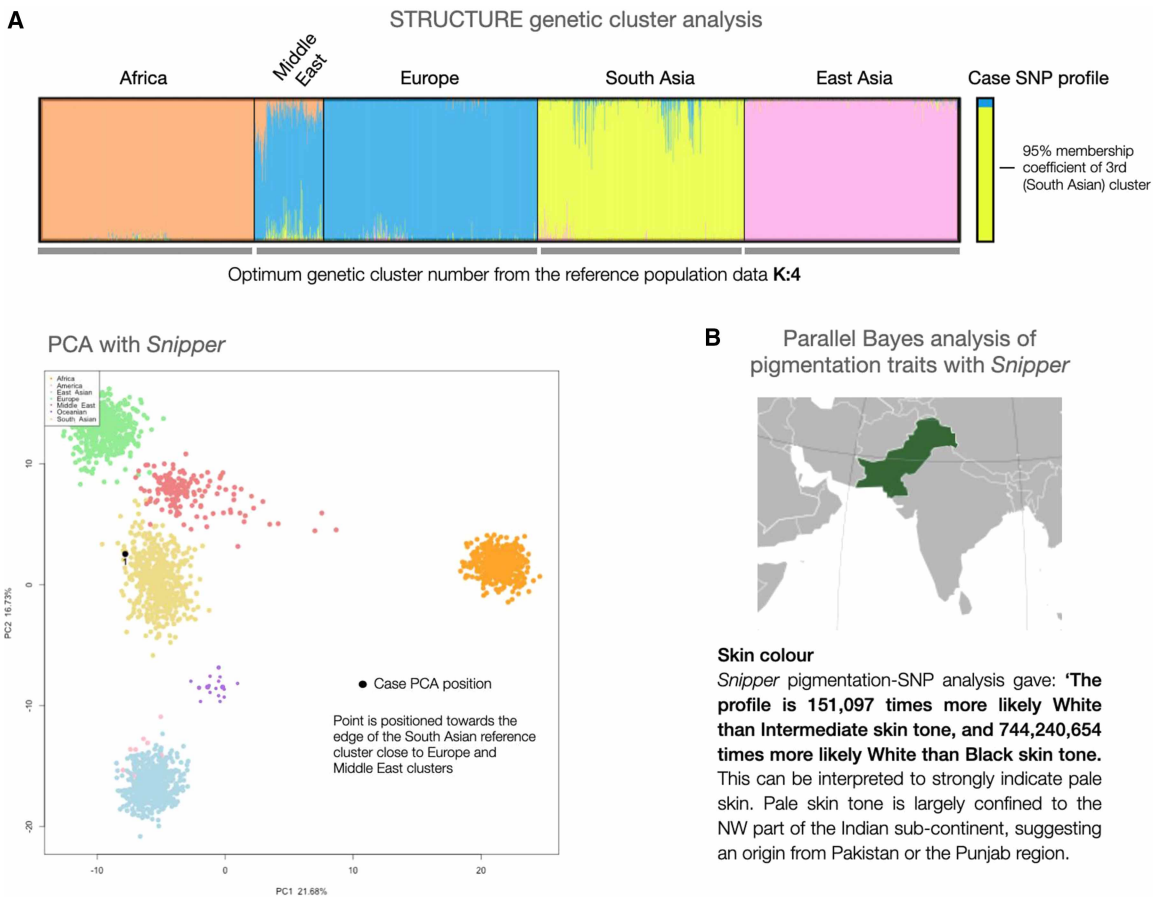


Figure 4. Differentiating closely related populations in a criminal investigation.

(**A**) PCA and STRUCTURE population data analysis using 144 AIM-SNPs in an MPS assay, applied to the investigation of a murder of a European in a country where Middle East and South Asian ancestries form the principal parts of the region's demographic profile. DNA recovered from the crime scene was inferred to be from a South Asian individual. In this case, PCA gave a more definitive distinction of Middle East and South Asian ancestries than STRUCTURE at K:4 or K:5 (data not shown). (**B**) Parallel pigmentation SNP tests on the crime scene DNA using Snipper Bayes likelihood ratio tests indicated the individual was extremely likely to have pale skin. This refined the ancestry inference reported to the investigators, which suggested the suspect originated from Pakistan or the Punjab region of India.

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this is obviously a product of too little informative variation data obtained from highly compromised DNA. The phenomenon of PCA displacement and LR reduction from marker drop-out is explored more fully in Phillips, 2015 (see Figures 4 and 5 in [52])

A murder investigation that differentiated closely related populations of origin

The criminal investigation of the murder of a European in a Middle East country with a large South Asian itinerant workforce, required an MPS-based analysis capable of distinguishing those three closely related population groups [27]. Figure 4 shows DNA found at the scene gave a clear signal of South Asian ancestry. Note the PCA gives the same triangle of European-African-East Asian clusters seen in Figure 3, with the other four, less divergent population clusters occupying the Euclidean space in between. STRUCTURE analyses did not distinguish Middle East and South Asian clusters at K : 5, so the risk of over-fitting was avoided, and the PCA made a more distinct separation of these populations, with the case SNP profile positioned on the edge (European side) of the South Asian reference cluster. The authors' laboratory always runs forensic appearance trait analyses [53] alongside ancestry tests and in this case, there was a degree of 'synergy' between both sets of results which helped refine the ancestry prediction. Pigmentation trait predictions in *Snipper* use identical Bayes LR analyses to ancestry inference and here the probability of pale skin tone was relatively high [54]. This was interpreted (along with the minor PCA deflection) to indicate a geographic origin from the NW region of South Asia, where pale skin predominates. A Pakistani suspect was eventually arrested, in part, based on this information. As in all criminal investigations where ancestry data has guided the search for suspects, identity was established with standard STR profiling, which confirms or excludes the presence of the suspect at the crime scene.

Summary

- The major developmental steps that have led to current forensic genetic ancestry analysis approaches are reviewed.
- The ancestry-informative marker sets and optimised capillary electrophoresis or massively parallel sequencing assays available to the DNA practitioner are compared.
- The range of statistical analysis regimes applicable to small-scale forensic ancestry-informative marker tests are described.
- Three forensic casework applications illustrate the complimentary population data analysis frameworks that can be combined to infer the ancestry of forensic DNA.

Competing Interests

The authors declare that there are no competing interests associated with the manuscript.

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Authors Contribution

The authors C.P. and M.d.l.P. shared responsibility for the planning, arrangement and writing of the manuscript. Both C.P. and M.d.l.P. have developed many of the AIM marker sets, as well as the analytical and statistical techniques described. C.P. and M.d.l.P. are part of a team applying the techniques described in the manuscript to routine investigative casework in their laboratory.

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Abbreviations

AIMs, ancestry informative markers; CE, capillary electrophoresis; EGDP, Estonian Biocentre Genome Diversity panel; HGDP-CEPH, Human Genome Diversity Project-Centre d'Etude du Polymorphisme Humain; LR, likelihood ratio; MDS, multi-dimensional scaling; MPS, massively parallel sequencing; PCA, principal component analysis; PCoA, principal coordinates analysis; SGDP, Simons Foundation Genome Diversity Project; SNP, single nucleotide polymorphism; STR, short tandem repeat.

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